On Data

A Consolidated Roadmap for the Pacific Arctic Shipping–Ice Dataset (2015–2020)

Appendix: the data backbone for *Melting Routes*

Data construction and exploration

May 25, 2026

Abstract

This appendix is the single, authoritative account of how the analytic dataset behind the report was built. It explains where every variable comes from, how four heterogeneous data products were placed on one common hexagonal–monthly grid, every addition and deletion of data along the way, and the exploratory analysis that characterises the result. It deliberately merges what were originally four separate technical notes—vessel traffic, sea-ice, meteorology, and distance-to-port—into a single pipeline, so that the data backbone of the study is coherent, reproducible, and auditable from raw download to final table. Throughout, the emphasis is on *decisions and their justification*: at each step we state not only what was done but why that option was preferred over the alternatives, and what limitation it leaves behind. Two issues receive particular attention because they propagate directly into later inference: the recoding of missing vessel counts as *structural zeros*, and the strong negative association between sea-ice concentration and air temperature, which we address at the data stage by constructing an ice-orthogonalised temperature covariate. We also localise exactly where, and why, the balanced panel loses a small number of rows during environmental extraction.

Note on figures. Figures whose captions are rendered as dashed placeholder panels are produced by the consolidated data script and should be dropped into `figs/` before the final build; the surrounding text fully describes each. The correlation figure (Fig. 8) is rendered from the actual assembled data.

Contents

1. Purpose and the analytic unit

1.1 What this document is, and why it exists

The central question of the study is whether, and how, monthly vessel-traffic intensity in the Pacific Arctic responds to contemporaneous environmental conditions. Answering it requires four very different data products to be made comparable: satellite Automatic Identification System (AIS) vessel traffic, satellite-derived sea-ice concentration, atmospheric reanalysis fields, and port geography. Each arrives in its own projection, resolution, and file format. Before any statistical question can be posed, these have to be harmonised onto one shared spatial and temporal support, cleaned of their idiosyncratic missing-value conventions, and joined into a single table. This roadmap documents that process end to end and is intended to stand alone in the appendix as the data backbone of the report: a reader who wants to know exactly what a row of the dataset means, and how it was produced, should find the complete answer here.

1.2 Why a hexagonal grid, an equal-area projection, and a monthly step

Every layer is summarised onto a fixed grid of 6,553 hexagonal cells in **EPSG:3338 (NAD83 / Alaska Albers)** at a **monthly** time step. Three design choices underpin the entire dataset, and each was made deliberately.

Hexagons rather than square pixels. A hexagonal tessellation has a single, constant distance between the centroids of adjacent cells in every direction, whereas a square raster has two (edge neighbours are closer than diagonal neighbours). This isotropy makes spatial neighbourhood structure cleaner and reduces orientation artefacts. Equally important from a practical standpoint, the vessel-traffic product is *natively* delivered on this hexagonal grid; adopting it as the master grid means the response variable is never resampled, and only the environmental layers—which are continuous fields and resample gracefully—are moved onto it.

An equal-area projection. In any geographic (longitude/latitude) coordinate system, a cell of fixed angular size shrinks in physical area as latitude increases. At Arctic latitudes this distortion is severe, and it would silently bias every area-weighted summary toward high-latitude cells. EPSG:3338 is an equal-area projection, so each hexagon represents the same physical area regardless of where it sits in the domain, and every cell contributes on equal footing to any spatial statistic.

A monthly grain. The AIS response is distributed as monthly aggregates. Matching the environmental layers to that same monthly grain produces a clean, balanced panel and, crucially, avoids manufacturing temporal precision the response simply does not have. The price of this choice—the loss of synoptic, storm-scale weather signal—is real and is acknowledged explicitly in §13.

Decision. Adopt the native hex grid (EPSG:3338) as the master spatial support and a monthly time step; resample only the continuous environmental fields onto it. The panel is therefore balanced with one row per hexagon per month, nominally $6{,}553 \times 72 = 471{,}816$ rows.

2. Data sources at a glance

Table 1 summarises the provenance of each layer. All four are documented, citable, public products accessed either through a maintained R package or a stable public file; no hand-curated or ad-hoc data enters the pipeline. The whole workflow is implemented as one annotated R script, and the ERA5 retrieval is performed from within R (via the `ecmwfr` interface to the Copernicus Climate Data Store) so that the pipeline never switches languages.

Layer	Product	Native CRS / res.	Access	Citation
Vessel traffic	N. Pacific & Arctic Marine Vessel Traffic Dataset (hex SpeedHex, AIS)	EPSG:3338, hex grid	72 monthly shapefiles	Kapsar et al. 2022b; 2023
Sea ice	NOAA/NSIDC passive-microwave sea-ice concentration CDR (`aice`)	EPSG:3413, 25 km	`icecdr` R pkg	Cavalieri et al. 1996; Lavergne et al. 2019
Meteorology	ERA5 monthly means: 2 m temperature, 10 m wind speed	0.25° lon/lat	`ecmwfr` / CDS	Hersbach et al. 2020
Ports	World Port Index (NGA) / Natural Earth ports	geographic (WGS84)	R pkg / public file	NGA WPI; Christensen et al. 2019

Table 1. Provenance of the four data layers.

3. Layer 1 — Vessel traffic (AIS)

3.1 Source, and what the product already contains

The response variable originates in satellite AIS messages—position reports broadcast by vessels and received by satellite and terrestrial stations. Turning billions of raw messages into a tidy hexagon-month table is a substantial engineering task in its own right, and it was performed upstream by Kapsar et al.: AIS messages were de-duplicated and quality-filtered, joined to a static vessel registry to obtain type and dimensions, segmented into daily trajectories, and intersected with the hex grid. We deliberately begin our pipeline *after* that work, taking the published product—72 monthly shapefiles, each containing the hex geometry and an attribute table of per-class traffic statistics—as our starting point. This avoids re-deriving (and potentially corrupting) a carefully validated AIS product, and it guarantees our response matches the one used in the reference literature.

Vessel classes follow a suffix convention that recurs throughout the dataset: `C` (cargo), `T` (tanker), `F` (fishing), `O` (other—icebreakers, tugs, research and passenger vessels), `L` ("long-fast": longer than 65 ft and faster than 10 kn), and `A` (all classes combined). Metric prefixes are `nMMSI` (count of unique vessels), `nOprD` (count of unique vessel-day combinations, an intensity measure), and `SOG`/`SOGsd` (mean and standard deviation of speed over ground).

3.2 Step 1.1 — Stack the 72 monthly files into one panel

Each monthly archive is unzipped, read as a spatial object, tagged with the calendar `date` implied by its file, and row-bound to the others. Because every month shares the same set of `hexID`s, stacking yields a balanced panel in which the same cell can be tracked through time—this is exactly the structure that later lets us separate variation *within* a cell over time from differences *between* cells.

```
read_speedhex <- function(zip_path) { # one month -> sf object
  shp <- unzip_to_temp(zip_path) # extract the .shp
  st_read(shp, quiet = TRUE) |>
    mutate(date = ymd(paste(year, month, "01", sep = "-")))
}
shipping_sf <- map_dfr(zip_files, read_speedhex) # -> 471,816 rows (6,553 hex x 72 months)
```

3.3 Step 1.2 — The response variable

The primary response is `nOprD_A`: total vessel-days for all vessel classes in a given hex-month. It is best understood as an *intensity*—bounded below by zero, dominated by zeros, and with a long right tail. It is not a clean Poisson count, which is why the modelling stage adopts a Tweedie-type response; that choice, however, belongs to the modelling appendix and is mentioned here only so the reader understands why `nOprD_A` is the column everything else is built around.

3.4 Step 1.3 — Structural zeros: verify before recoding

This is the single most consequential data-handling decision in the project, so it is worth stating carefully. In the raw attribute table, a count cell is left missing (`NA`) when no vessel of that class was recorded in that hexagon that month. Interpreted correctly, such a cell is a *true, observed zero*: the absence of traffic is itself the datum. Interpreted carelessly—if a missing value were actually a data gap, such as a month of degraded AIS reception—recoding it to zero would fabricate an observed absence where there was really no observation, and would contaminate every statistic about the zero process, which is central to the study.

Decision. Do not recode blindly. Before converting anything, the pipeline verifies that missingness is structural, by inspecting (i) the per-month `NA` fraction of `nOprD_A` and `nMMSI_A`; (ii) the per-class `NA` fractions; and (iii) that no entire month or layer is missing (which would signal a reception gap rather than genuine absence of traffic). Only after these checks pass do we recode `NA`→ 0 *for count columns only.* Speed columns (`SOG_*`, `SOGsd_*`) are left as `NA`, because the mean speed of zero vessels is undefined and a zero there would be a falsehood.

```
count_vars <- grep("^(nMMSI|nOprD)", names(shipping_sf), value = TRUE)
# (only after the three structural-zero checks above have passed)
shipping_sf <- shipping_sf |>
  mutate(across(all_of(count_vars), ~ replace_na(.x, 0))) # SOG_* deliberately left NA
```

The pipeline prints the number of cells recoded and the resulting zero proportion of `nOprD_A`, both overall and within ice-concentration bins; these appear in the exploratory section (§11). The proportion is high, and that fact is the empirical justification for using a zero-aware response distribution later.

3.5 Step 1.4 — Vessel scope is a documented filter, not a footnote

When the main analysis is run over a subset or grouping of vessel classes (for example, focusing on the classes relevant to commercial transit), that restriction is a genuine *data filter*: it changes the population to which every subsequent result applies. Such a choice is therefore recorded here and carried into the variable dictionary, rather than being left implicit in the narrative of the main report. The dataset retains all per-class columns so that any grouping can be reconstructed and audited.

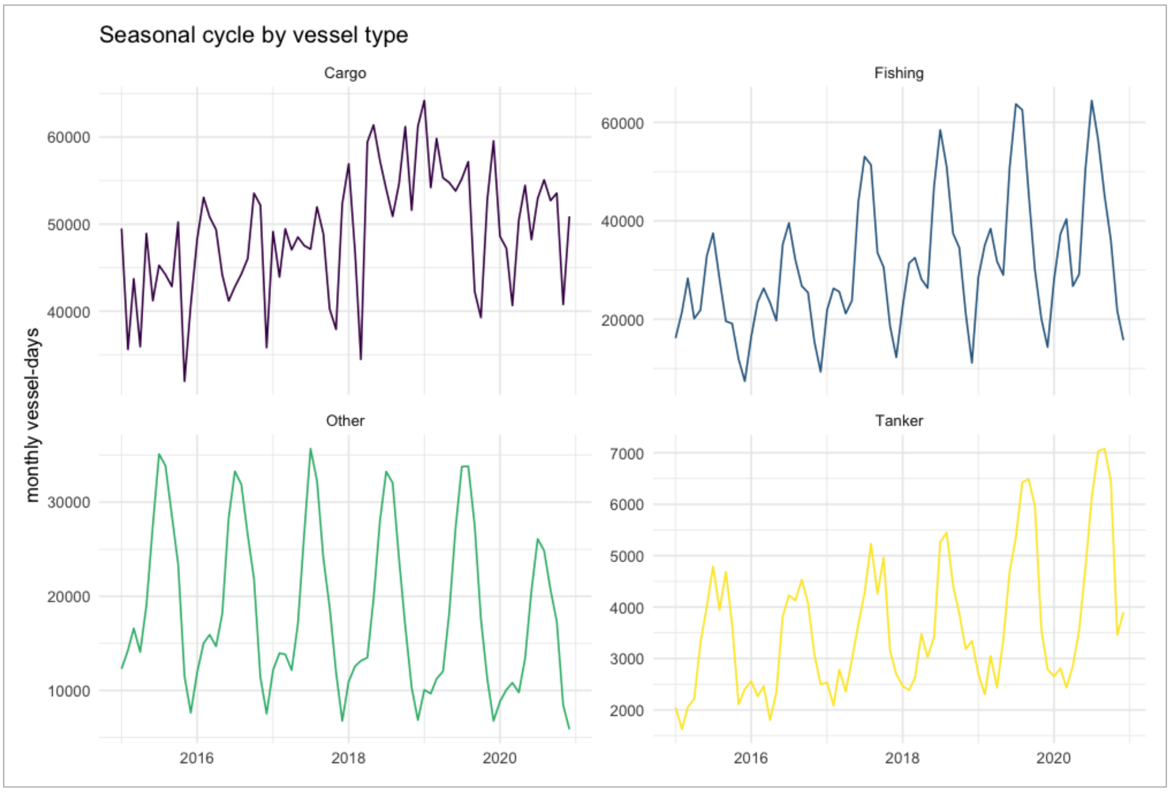

Figure 1. Seasonal cycle of monthly vessel-days by vessel class, 2015–2020. The pronounced summer peak is the empirical basis for an explicit seasonal term in the model. `nOprD_A`.

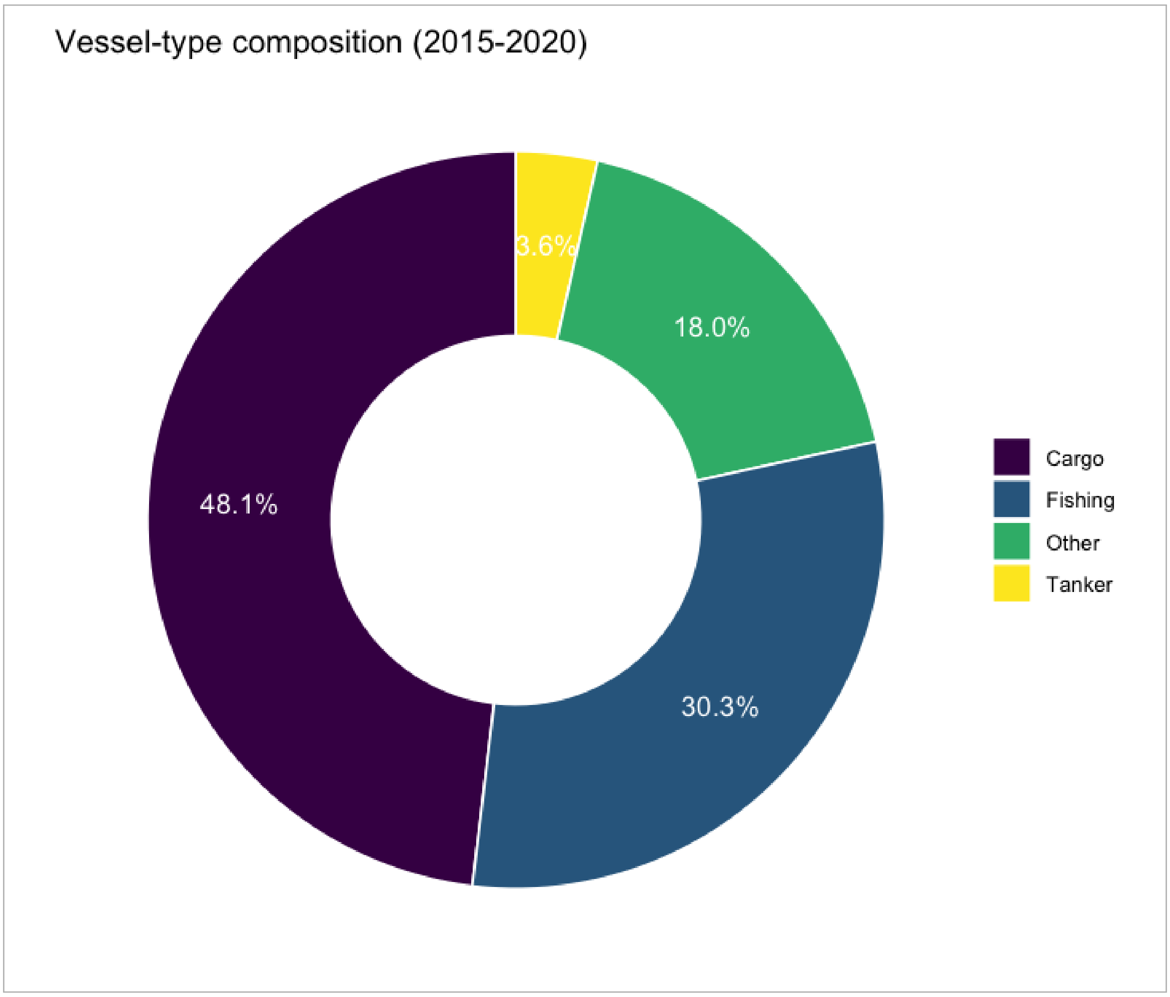

Figure 2. Overall composition of traffic by class (right), 2015–2020. The composition shows which classes dominate the all-vessel response `nOprD_A`.

4. Layer 2 — Sea-ice concentration

4.1 Source, and why it is the primary predictor

Sea-ice concentration is the dominant physical control on Arctic navigation: it determines whether a route is passable at all, governs achievable speed, and shapes the spatial pattern of traffic. It is therefore the central environmental covariate of the study. We obtain the NOAA/NSIDC passive-microwave sea-ice concentration Climate Data Record through the `icecdr` R package, which exposes the variable `aice` (a fraction between 0 and 1) and offers a single-command, reproducible download—avoiding manual interaction with the data server and matching the satellite record used in the reference literature. The pipeline pins and prints the exact product and version so that any ambiguity between the Climate Data Record and the related Sea-Ice-Index product is resolved on the record.

4.2 Step 2.1 — Download and assign the coordinate system

The downloaded NetCDF does not embed its projection. This is a known pitfall: if the CRS is left unset, the subsequent reprojection either fails or—worse—succeeds silently with wrong coordinates.

Decision. Assign the NSIDC North Polar Stereographic CRS (EPSG:3413) explicitly and immediately after loading, before any spatial operation.

```
ice_file <- cdr_arctic_monthly(c("2015-01-01","2020-12-01"),
                             variables = "aice", version = 5, dir = "ice_data")
ice_stack <- rast(ice_file)
crs(ice_stack) <- "EPSG:3413" # mandatory: not embedded in the raw file
```

4.3 Step 2.2 — Reproject and extract an area-weighted mean per hexagon

The ice stack is warped from its native 25 km polar-stereographic grid onto the hex CRS (EPSG:3338) using **bilinear** interpolation, and then summarised to each hexagon by an **area-weighted** mean computed with `exactextractr`, which knows the exact fraction of every 25 km cell that falls inside each hexagon.

Decision. Use bilinear (not nearest-neighbour) reprojection because ice concentration is a smoothly varying continuous field, so interpolation is appropriate and avoids artificial blockiness. Use area-weighted extraction (not centroid sampling) because a centroid value discards the within-hexagon distribution and oversmooths precisely the ice-edge region that the study most cares about; the area-weighted mean correctly handles hexagons only partially covered by a raster cell.

```
ice_3338 <- project(ice_stack, "EPSG:3338", method = "bilinear")
for (m in 1:nlyr(ice_3338)) {
  vals <- exact_extract(ice_3338[[m]], hex_grid, fun = "mean") # area-weighted
  ice_df <- bind_rows(ice_df,
                    data.frame(hexID = hex_grid$hexID,
                               date = time(ice_3338)[m],
                               ice_conc = vals))
}
```

4.4 An aggregation caution that matters for any threshold

Because the hex value is an area-weighted average, a hexagon straddling the ice edge receives an *intermediate* concentration that may physically correspond to “half open water, half pack ice” rather than a uniform moderate cover. Any threshold later read off `ice_conc` is therefore a threshold on this *aggregated* quantity, not on the ice a single vessel actually meets. Figure 3 makes the effect of aggregation directly visible by placing the raw raster beside the hex field for the same month.

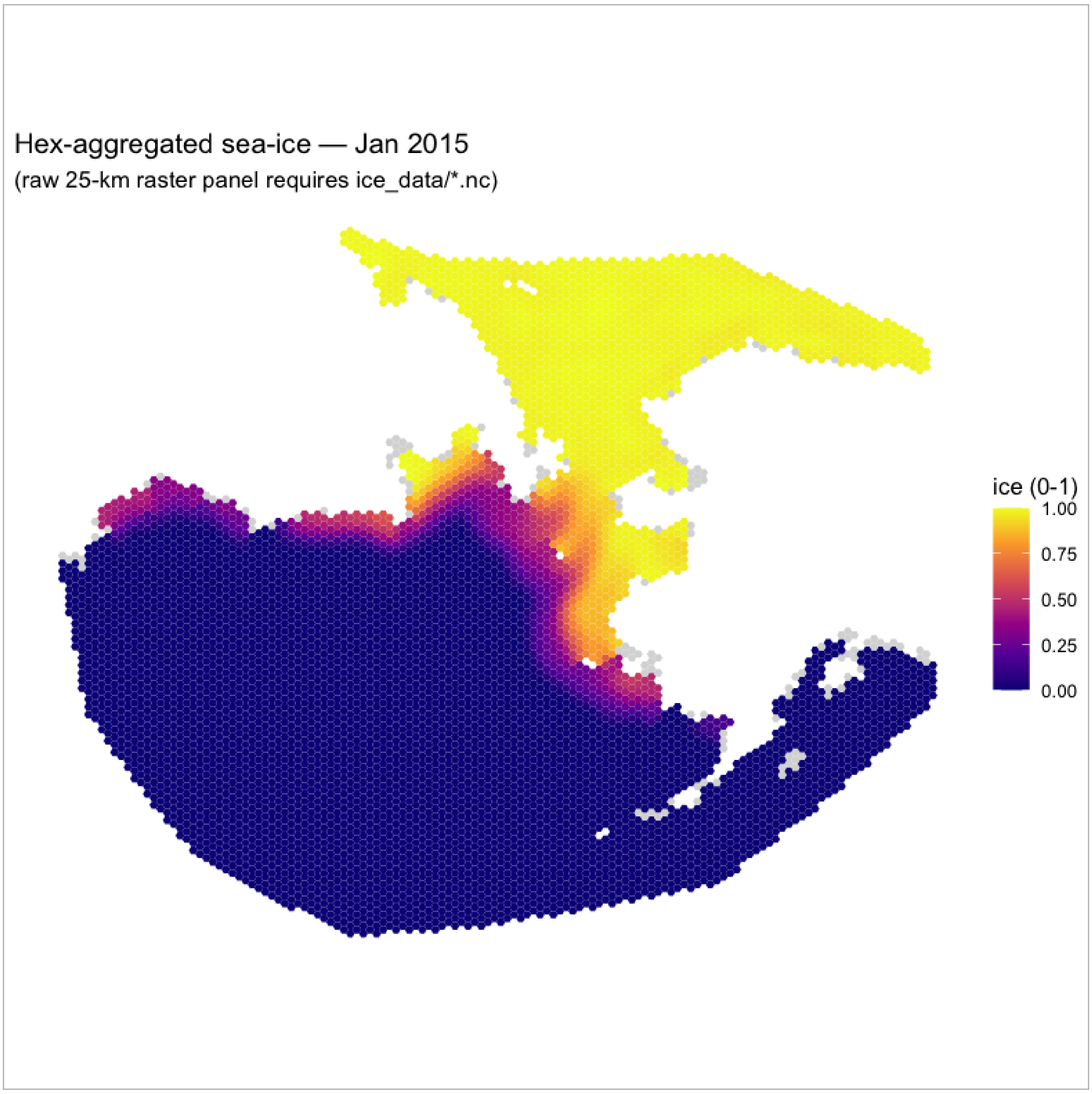

Figure 3. Sea-ice concentration for January 2015: the original 25 km NSIDC raster shown in EPSG:3413. The hex view necessarily smooths the marginal ice zone—an unavoidable consequence of any zonal averaging, and the reason threshold statements must be framed in terms of the aggregated variable.

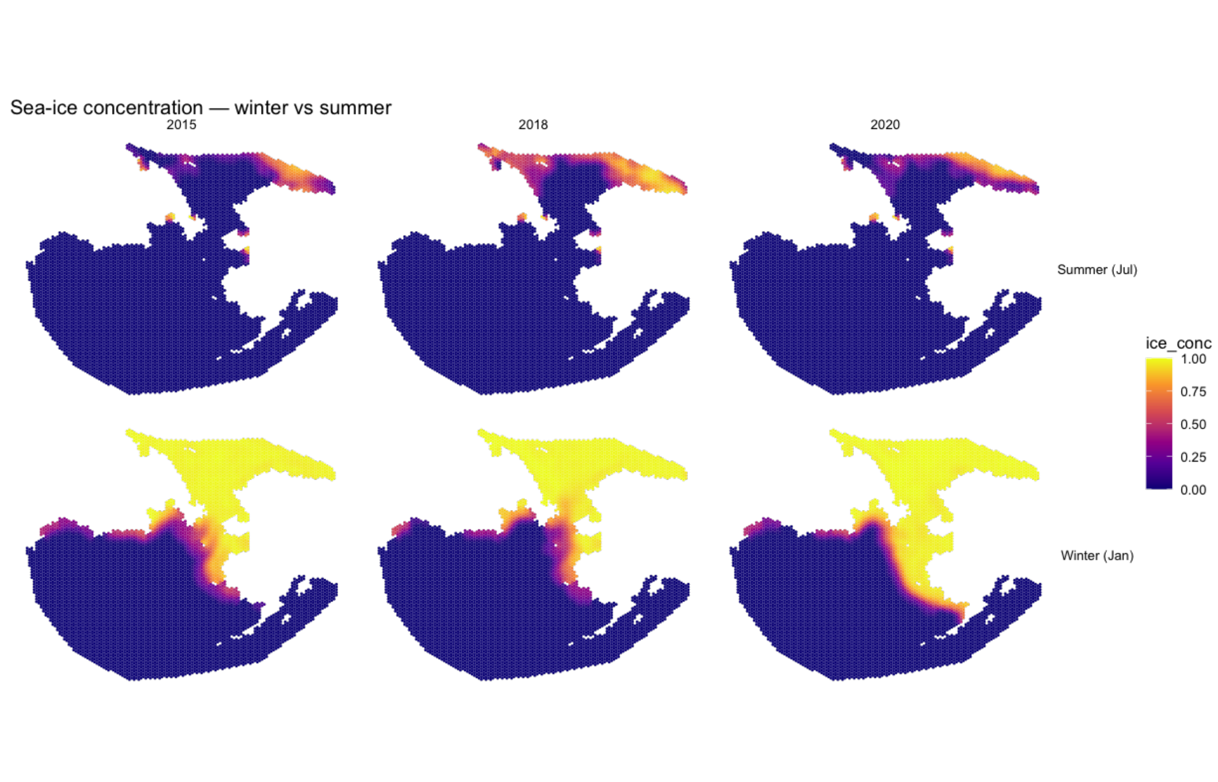

Figure 4. Hex-level ice concentration for selected winter and summer months across the study period. The maps reproduce the expected seasonal advance (winter) and retreat (summer) of the ice edge, serving as a qualitative validation that the reprojection and extraction behave correctly.

4.5 Step 2.3 — Merge with traffic

The per-hexagon ice table is joined to the cleaned traffic attributes on the keys `(hexID, date)` with a left join keyed on the traffic panel. Because the join is keyed on the response, it can only retain or annotate existing rows; it cannot multiply them. Missing ice values are introduced only where a hexagon has no overlapping ice data, and those cases are tracked precisely in §8.

5. Layer 3 — Meteorology (ERA5)

5.1 Source, and why these two variables

Beyond ice itself, the meteorological variables most consistently identified as Arctic maritime risk factors are near-surface wind and temperature. We therefore take ERA5 monthly means of **2 m air temperature** and **10 m wind speed** from the Copernicus Climate Data Store. ERA5 is the fifth-generation ECMWF reanalysis, extensively validated against in-situ Arctic observations and widely used in polar studies; pre-computed monthly means are available directly for our window.

Decision. Use the published monthly means rather than downloading hourly fields and averaging them ourselves. This avoids transferring and processing thousands of hourly layers, and it guarantees our aggregation matches the standard product. The cost—an inability to represent sub-monthly weather events—is accepted and documented below.

5.2 Step 3.1 — Retrieve in a single request, from within R

A single request retrieves both variables for the whole period over a bounding box that spans the entire grid, including both sides of the 180° meridian. The Copernicus area argument is ordered

[North, West, South, East]; an earlier version that mis-ordered these produced a wrong spatial subset, so the order is fixed and commented.

```
# retrieved via ecmwfr (CDS API) so the pipeline stays single-language
request <- list(dataset = "reanalysis-era5-single-levels-monthly-means",
  product_type = "monthly_averaged_reanalysis",
  variable = c("2m_temperature","10m_wind_speed"),
  year = 2015:2020, month = sprintf("%02d", 1:12), time = "00:00",
  area = c(80, 180, 40, -180), # N, W, S, E -- spans both sides of the dateline
  format = "netcdf")
```

5.3 Step 3.2 — Reproject, extract, and convert units

The temperature and wind stacks are processed with exactly the same machinery as the ice layer: bilinear reprojection to EPSG:3338 followed by area-weighted extraction per hexagon.

Decision. Use identical reprojection and extraction for all environmental layers, so that every predictor shares the same spatial support and the same averaging rule—this consistency is what makes the predictors directly comparable in a single model.

Temperature is converted from Kelvin to Celsius for interpretability; because this is a linear shift it has no effect on any smooth term estimated later. Dates are derived from the NetCDF time axis (seconds since 1970-01-01), never inferred from file names.

```
temp_3338 <- project(era5[[1:72]], "EPSG:3338", method = "bilinear")
wind_3338 <- project(era5[[73:144]], "EPSG:3338", method = "bilinear")
# air_temp = exact_extract(temp_3338[[m]], hex_grid, "mean") - 273.15 # Kelvin -> Celsius
# wind_speed = exact_extract(wind_3338[[m]], hex_grid, "mean") # m/s
```

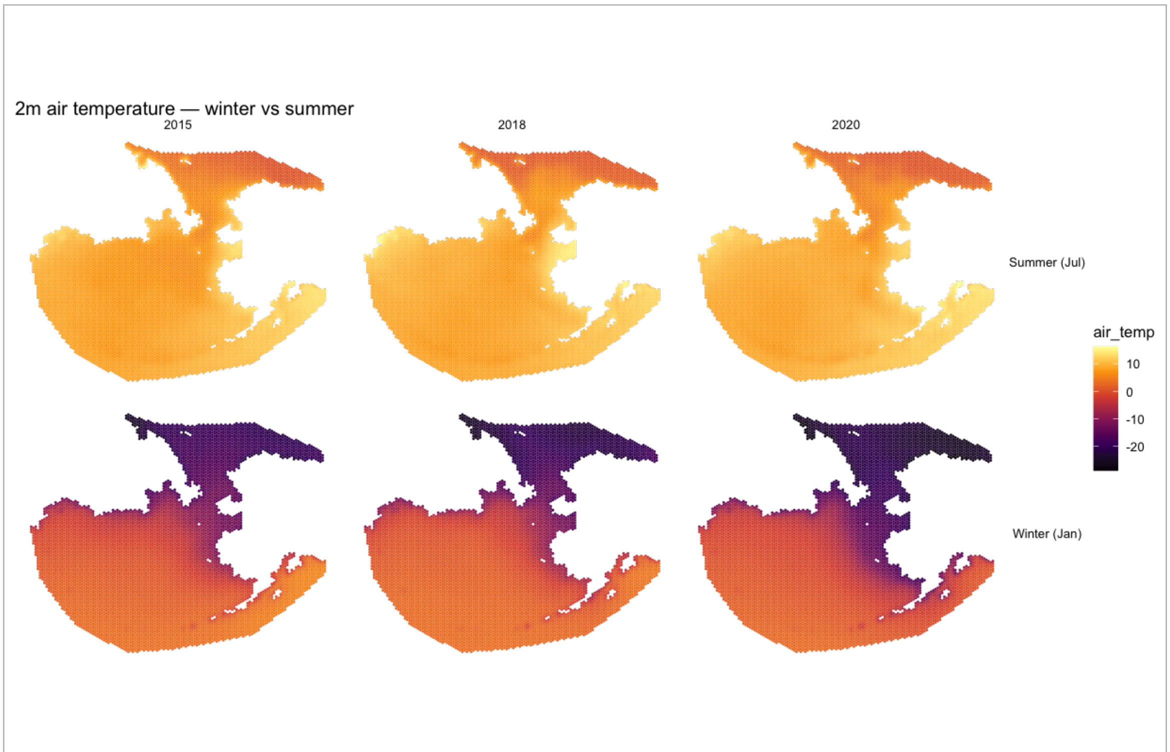

Figure 5. Spatial summaries of the meteorological layers: mean 2 m air temperature. This summary serves as a sanity check that reprojection and extraction reproduce the known climatology (a strong winter north–south temperature gradient; elevated winds in the Bering Sea).

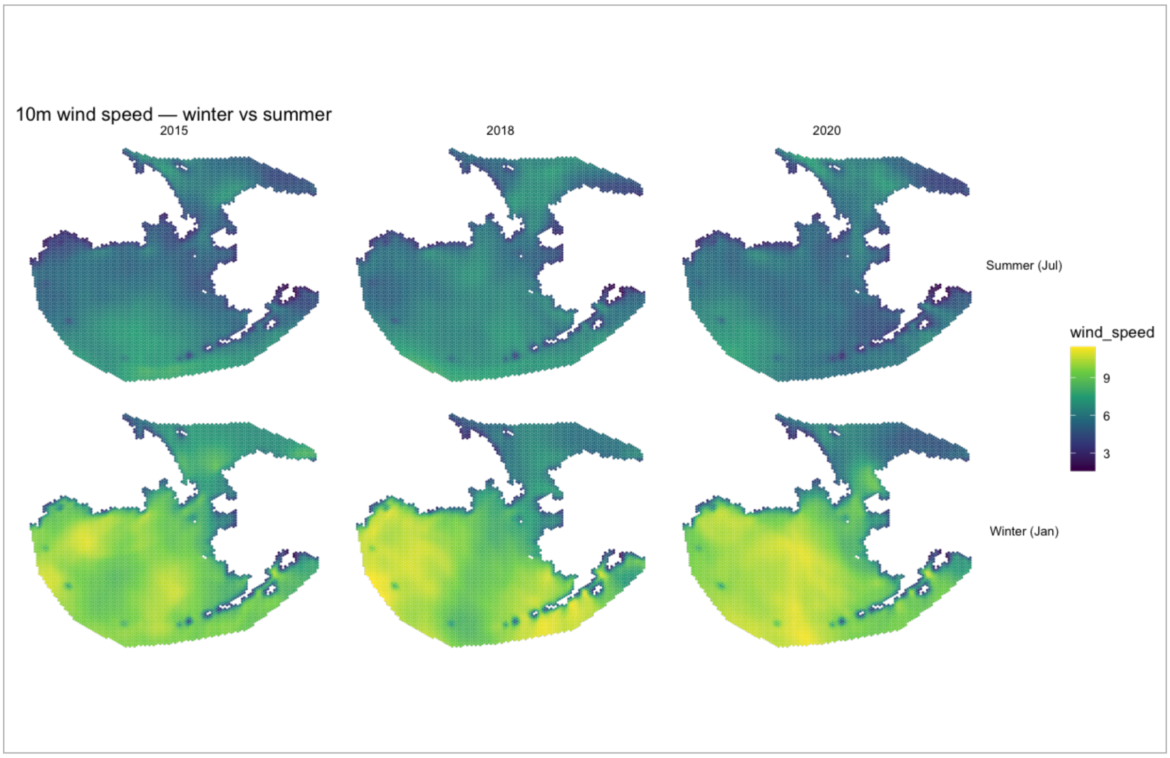

Figure 6. Spatial summaries of the meteorological layers: mean 10 m wind speed (right, with high-ice cells overlaid). This summary serves as a sanity check that reprojection and extraction reproduce the known climatology (a strong winter north–south temperature gradient; elevated winds in the Bering Sea).

5.4 The monthly-mean caveat

Monthly means cannot represent storms, cold snaps, or other synoptic events. The wind and temperature variables therefore enter the analysis as *seasonal and large-scale controls*, and the study makes no claim to capture event-scale weather effects. The justification for including these variables is framed accordingly: they account for broad climatological context, not short-lived risk episodes.

6. Layer 4 — Distance to nearest port

6.1 Why it is included, and why it had to be rebuilt

Commercial shipping is fundamentally port-to-port, and remoteness from a port raises both the practical difficulty and the insured risk of a voyage: rescue and ice-breaker support take longer to arrive, so operators accept tighter ice limits far from a safe haven, and traffic density falls with distance from ports even where the ice itself is thin. Distance-to-port is thus a parsimonious proxy for several correlated remoteness effects at once, which is exactly why it is worth including.

The variable's *motivation* was never in doubt; its earlier *implementation* was. An initial version computed distance against a short, hand-typed list of ports—two of which lay outside the study frame—using a distance metric that was never made explicit. That is neither reproducible nor faithful to the methodology it cited (Christensen et al., who derive distance-to-port from the World Port Index).

Decision. Rebuild the layer from a documented source with an explicit selection rule and a single, stated distance metric.

6.2 Step 4.1 — A documented port source and a reproducible selection rule

Ports are taken from a documented public source—the NGA World Port Index, or Natural Earth ports via the `rnaturalearth` package—and selected by a *rule*, not by hand: all ports falling within the study bounding box (equivalently, above a stated latitude threshold). The full list of ports actually used—name, country, and coordinates—is printed by the pipeline and reproduced in the report, so the selection is fully auditable and can be reproduced by anyone.

6.3 Step 4.2 — One distance metric, one CRS, stated plainly

For each hexagon centroid we compute the distance to the nearest selected port using a single, explicit method. We use **geodesic great-circle distance** (via `sf::st_distance` on geographic coordinates), which is unambiguous and correct at high latitude; if a planar distance is preferred for speed, both the centroids and the ports are first projected to EPSG:3338 and that fact is recorded. The metric and CRS are never left to assumption.

```
ports <- load_world_port_index() |> filter(in_study_box) # documented source + explicit rule
centroids <- st_centroid(hex_grid) # EPSG:3338
d_km <- apply(st_distance(centroids, ports), 1, min) / 1000 # geodesic, nearest port
port_dist <- d_km
log_port_dist <- log1p(d_km) # documented transform
```

6.4 Step 4.3 — An endogeneity caveat we carry openly

There is a circularity worth naming: ports tend to be located where traffic already concentrates, so `port_dist` partly encodes the very thing the study seeks to explain. The variable is retained as a useful remoteness proxy, but the report flags that part of its apparent explanatory power is mechanical rather than causal, and recommends checking that the ice signal is stable whether or not `port_dist` is included. We do not attempt to "solve" this at the data stage; we document it so that it is interpreted honestly downstream.

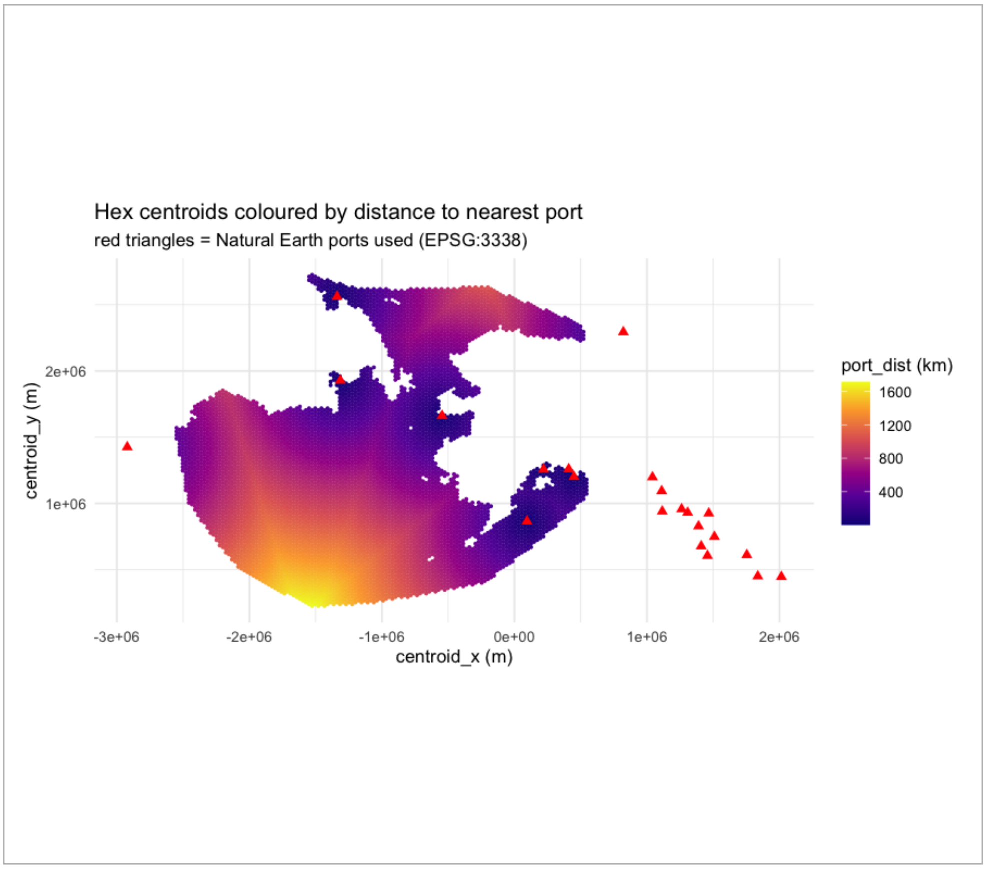

Figure 7. Hexagon centroids coloured by geodesic distance to the nearest selected port (EPSG:3338), with the selected ports overlaid. The companion panel shows the distribution of `port_dist`. This map also doubles as a check that the port selection rule produced a sensible, study-relevant set of ports.

7. Assembly — building the analytic table

With all four layers on the common grid, the analytic table is assembled by a sequence of left joins, always keyed on the traffic panel so that the join can only retain or annotate rows, never duplicate them:

```
df <- cleaned_shipping |>
  left_join(ice_conc, by = c("hexID","date")) |>
  left_join(air_temp, by = c("hexID","date")) |>
  left_join(wind_speed, by = c("hexID","date")) |>
  left_join(port_dist, by = "hexID") # port_dist is time-invariant
```

A printed **row/hex ledger** records, after every read, join, recode, and filter, the number of rows, the number of unique `hexID`s, the number of unique dates, and the `NA` count for each key variable. The purpose of the ledger is simple but important: nothing is ever dropped silently, and any change in the size or completeness of the table can be traced to the exact step that caused it.

8. Where the panel loses rows, and why

The complete balanced panel contains 6,553 $\times$ 72 $=$ 471,816 rows. The analytic table used for the univariate checks, however, contains N = 461,448 rows. The arithmetic is exact and worth following: 461,448/72 = 6,409 hexagons, so 6,553 − 6,409 = 144 hexagons—equivalently 144 $\times$ 72 = 10,368 rows—are absent.

Decision. Treat this loss as something to be *located and explained*, not quietly tolerated.

The loss does not arise in the traffic stack, which is complete by construction. It is introduced during the **environmental extraction**: a hexagon lying outside the coverage of a raster returns `NA` from `exact_extract`. Two mechanisms can do this—an ice cell that is masked or beyond the product's spatial extent, or an ERA5 cell beyond the requested bounding box or at the dateline/pole edge—and the 144 affected hexagons are edge cases of exactly this kind. Because a row with a missing predictor cannot enter the analysis, these rows are removed; but the removal is now an explicit, instrumented step rather than an implicit `na.omit`.

Concretely, the pipeline reports, per hexagon, how many cells return `NA` from the *ice* extraction versus the *air* extraction, attributes each of the 144 hexagons to the responsible layer, writes the dropped `hexIDs` to `dropped_hexes.csv`, and prints the count and cause to the log. *The exact split between the ice and air layers is reported by the current run*[run] and is inserted here once available. The governing principle is fixed regardless of the split: no row is deleted without a documented cause, a count, and an identification of which environmental layer drove it.

9. Derived variables and transformations

Every constructed column is documented with its definition, its purpose, and any caveat it carries.

9.1 Ice-orthogonalised temperature (`air_temp_resid`)

Sea-ice concentration and air temperature are strongly anti-correlated (§10): physically, cold air and heavy ice go together. This makes it difficult for any additive model to attribute effects to ice and temperature separately, because the two predictors carry overlapping information. To give the modelling stage the *option* of isolating the part of temperature that varies independently of ice, we construct an ice-orthogonalised temperature: the residual of temperature after its ice-explained component has been removed.

```
# This is a DATA TRANSFORMATION used only to construct a covariate -- NOT the analysis model.
fit_t <- gam(air_temp ~ s(ice_conc, bs = "ps"), data = df) # univariate helper regression
df$air_temp_resid <- residuals(fit_t) # ice-orthogonalised temperature
```

Decision. Construct `air_temp_resid` as a derived covariate, and treat the helper regression of temperature on ice strictly as a data-preparation step, never as a model of any outcome.

The pipeline reports the fraction of temperature variance explained by ice and plots `air_temp_resid` against `ice_conc` to confirm that the residual is, by construction, uncorrelated with ice. Downstream work can then choose between raw temperature and its ice-orthogonalised form. The trade-off is explicit: `air_temp_resid` removes the collinearity but is harder to interpret physically, and that limitation is stated wherever the variable is used.

9.2 Other derived columns

- `ice_free_flag` $= \mathbf{1}\{$`ice_conc` $< 0.15\}$. The conventional "open water" threshold, retained for description and for comparison with the literature, not as a modelling cut.

- `post2017`. A pre/post-1 January 2017 indicator marking the entry into force of the Polar Code. It is labelled honestly as a *step indicator*: a single step will absorb any shift in traffic after 2017 regardless of its true cause, so it is described as a pre/post marker rather than as "the Polar Code effect."

- `centroid_x`, `centroid_y`. Hexagon centroid coordinates in EPSG:3338, used for spatial description and for the distance-to-port computation. The plotting CRS (EPSG:3413) is kept deliberately distinct from the analysis CRS to avoid confusion.

- `time_index` $= 12(\text{year} - 2015) + \text{month}$. An integer month counter, convenient for describing the long-run trend over the six years.

10. Collinearity among the predictors

The pairwise Pearson correlations among the response and the three continuous predictors, computed on the assembled table ($N = 461{,}448$), are shown in Fig. 8. The values are modest except for one: `ice_conc` and `air_temp` correlate at -0.781. Ice and temperature therefore carry largely the same information, a relationship that is physically expected rather than surprising.

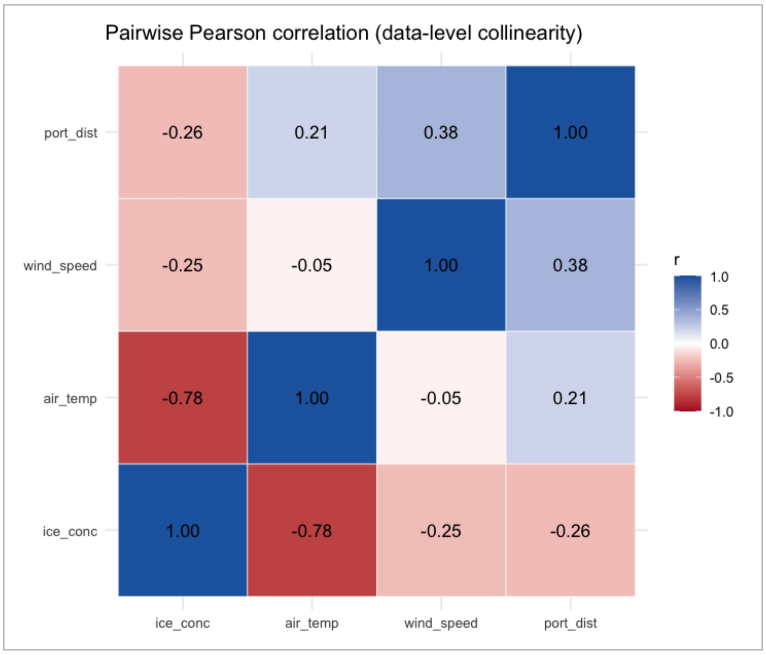

Figure 8. Pairwise Pearson correlations among the key variables, computed on the assembled analytic table. The dominant feature is the strong inverse relationship between ice concentration and air temperature (-0.781); the response `nOprD_A` is only weakly linearly related to any single predictor, which is itself informative (§11). *This figure is rendered from the actual data.*

Two consequences follow, and are recorded here. First, this strong association is the direct motivation for the ice-orthogonalised temperature covariate of §9. Second, and stated plainly: what is reported here is *collinearity at the data level*—a property of the raw variables. Its model-stage analogue, *concurvity* (the degree to which one smooth function can be approximated by a combination of the others), is a property of the fitted additive model and is therefore measured and discussed in the modelling appendix, not here. This roadmap's responsibility is to surface the evidence and to supply the orthogonalised covariate that lets that later discussion proceed on a sound footing.

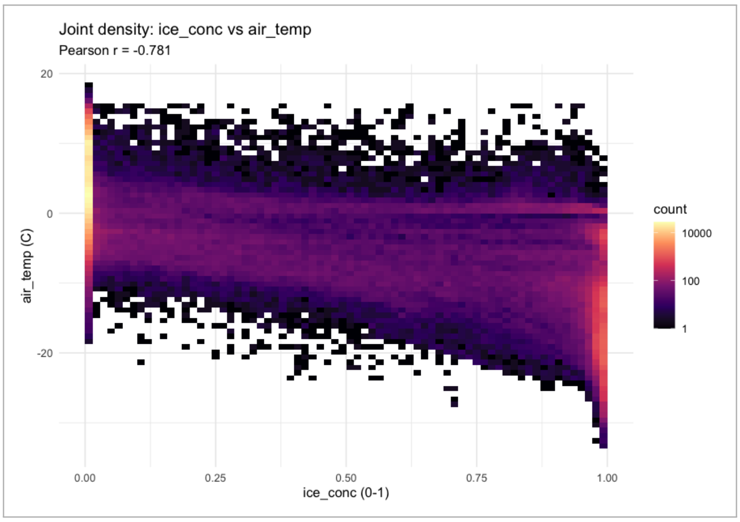

Figure 9. Joint density of `air_temp` and `ice_conc` (two-dimensional bins, logarithmic colour scale). The tight inverse band is the source of the -0.781 correlation and the concrete reason an ice-orthogonalised temperature is constructed.

11. Exploratory analysis of the assembled dataset

The exploration serves two purposes at once: to acquaint the reader with the shape and quirks of the data, and to expose the features that drive the modelling choices made later.

11.1 The zero process

The single most informative descriptive object is the **empirical proportion of zero-traffic hexagons as a function of ice concentration** (Fig. 10). This curve is the data-level analogue of an operational accessibility relationship: it shows the concentration at which the chance of *any* traffic collapses, without assuming any model. It is complemented by the proportion of zero-traffic cells by month and year (Fig. **??**), which reveals zero-inflation that is both pervasive and strongly seasonal—high in winter, lower in late summer. Together these motivate a response distribution that can accommodate a large, structured mass at zero.

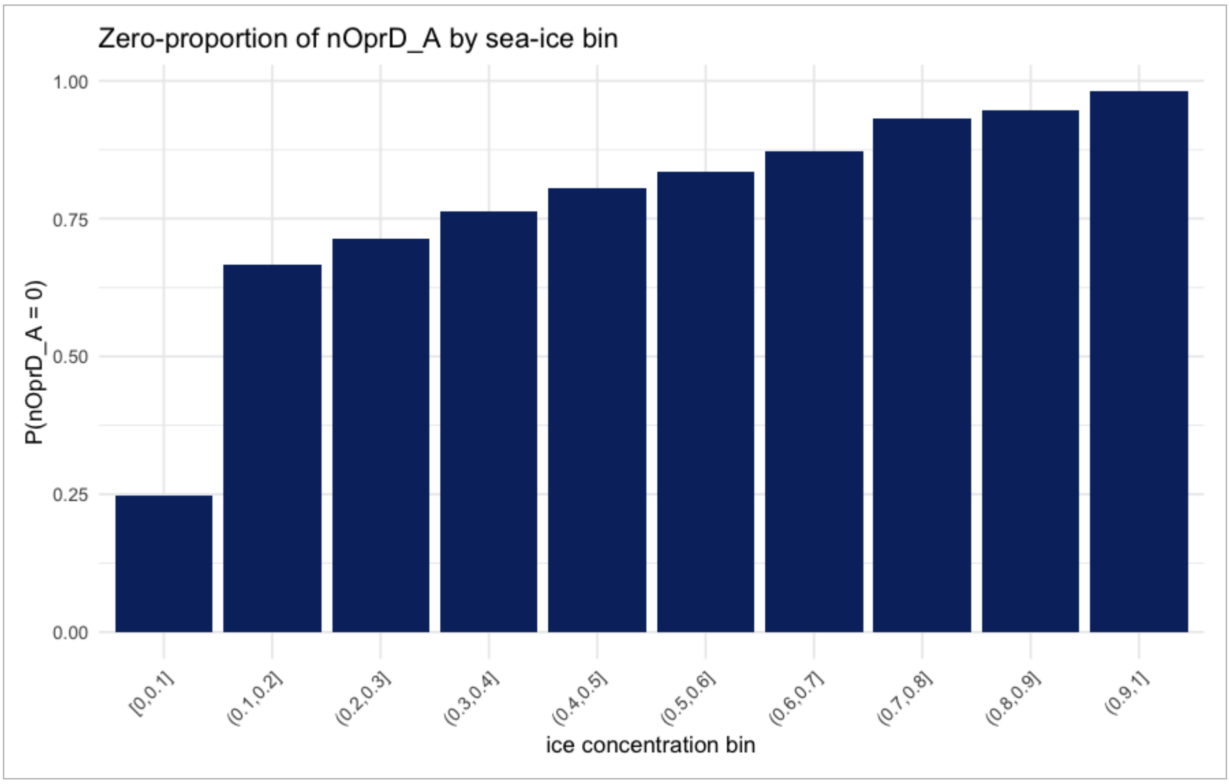

Figure 10. Empirical proportion of hexagons with zero `nOprD_A` within bins of ice concentration. The data-level accessibility curve: where it rises steeply marks the ice regime at which traffic effectively ceases.

11.2 The response distribution

Histograms of `nOprD_A`—across all hex-months, and conditional on positive traffic—show a sharp spike at zero followed by a long right tail (the maximum observed value is approximately 2,067 vessel-days). This shape, not any linear summary, is what the modelling response must reproduce (Fig. 11).

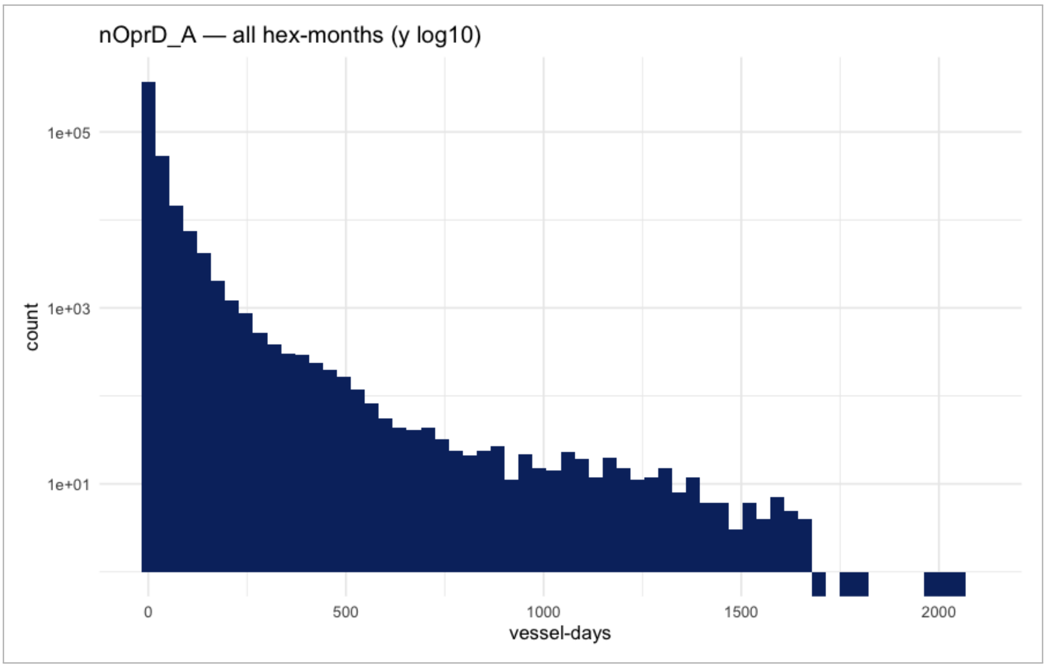

Figure 11. Distribution of `nOprD_A`: across all hex-months (the spike at zero) and conditional on positive traffic (the long right tail). Together they characterise a zero-inflated, heavy-tailed intensity.

11.3 Why the naive linear view is reported only as a warning

For completeness the pipeline records simple linear regressions of `nOprD_A` on each predictor. They are statistically "significant" yet explanatorily empty: the coefficient of determination is $R^2 \approx 0.017$ for ice, 0.023 for temperature, and 0.002 for wind, with residuals ranging from roughly -19 to $+2{,}067$.

Decision. Report these models *only* as evidence that ordinary linear regression is the wrong tool for a spike-at-zero, heavy-tailed response; their coefficients are not to be interpreted.

Their near-zero explanatory power, far from being a disappointment, is itself a useful finding: it tells us in advance that the response is governed by nonlinearity, spatial structure, and seasonality rather than by any simple linear effect.

11.4 Spatial and seasonal structure

The per-hexagon mean-traffic maps (Fig. **??**) and the seasonal-cycle figure (Fig. 2) together show that most of the explainable structure in the response is spatial and seasonal. This is important context to carry into the modelling stage: it warns, before any model is fit, that the *marginal* contribution of ice—once space and season are accounted for—may be modest, and that the honest task is to quantify that contribution rather than to assume it.

12. Final dataset — variable dictionary

The final product is a balanced panel with one row per `hexID` × month, spanning 2015-01 through 2020-12. Its columns are catalogued in Table 2.

Variable	Type	Definition / notes
`hexID`	key	Hexagon identifier (EPSG:3338 grid).
`date`, `year`, `month`	key	Month start; temporal index.
`nOprD_A`	numeric	Primary response: all-vessel vessel-days. Per-class `nOprD_*` retained.
`nMMSI_*`	numeric	Unique-vessel counts by class (count; `NA` recoded to 0).
`SOG_*`, `SOGsd_*`	numeric	Speed mean / s.d. by class; kept `NA` where no traffic.
`ice_conc`	numeric	Area-weighted sea-ice concentration, 0–1 (primary predictor).
`air_temp`	numeric	Area-weighted 2 m air temperature (°C).
`air_temp_resid`	numeric	Ice-orthogonalised temperature (residual of `air_temp` on `ice_conc`).
`wind_speed`	numeric	Area-weighted 10 m wind speed ($\mathrm{m\,s^{-1}}$).
`port_dist`	numeric	Geodesic distance (km) to nearest selected port; `log_port_dist` $= \log(1 + \cdot)$.
`ice_free_flag`	binary	$\mathbf{1}\{$`ice_conc` $< 0.15\}$ (descriptive).
`post2017`	binary	Pre/post 2017-01-01 step indicator.
`centroid_x/y`	numeric	Hexagon centroid, EPSG:3338.
`time_index`	integer	$12(\text{year} - 2015) + \text{month}$.

Table 2. Variable dictionary for the final analytic table.

13. Data-quality cautions and limitations

No dataset is free of caveats, and stating them plainly is part of making the backbone trustworthy.

1. **AIS scope.** The traffic product reflects AIS-reporting commercial and larger non-military vessels; military and some small craft are absent. The word "traffic" should be read in that specific sense throughout the report.

2. **Ice-edge aggregation.** Area-weighted hex means smooth the marginal ice zone, so any threshold expressed in `ice_conc` is a threshold on the aggregated quantity, not on vessel-scale ice (§4).

3. **Reanalysis and monthly means.** ERA5 blends a physical model with observations and, as monthly means, masks synoptic events; the meteorological variables are large-scale controls only (§5).

4. **Port endogeneity.** `port_dist` partly encodes where traffic already concentrates, so part of its effect is mechanical rather than causal (§6).

5. **Observational status and reverse causation.** Ships actively avoid ice, so `ice_conc` and traffic are jointly determined. Every relationship documented here is an *association*, never a causal effect.

6. **Spatial and temporal dependence.** Neighbouring hexagons and repeated observations of the same cell are correlated. This dependence is a structural property of the panel that the modelling stage must address explicitly.

14. Reproducibility

The entire pipeline is a single annotated R script driven by a top-of-file configuration block that fixes all paths, the date range, the CRS codes, and the ERA5 bounding box in one place. Raw downloads are skipped if the files are already present; a seed is set for any stochastic step; and `sessionInfo()` together with key package versions is printed at the end so the computational environment is recorded. The intermediate products—`cleaned_shipping.rds`, `ice_concentration.rds`, `air_temp.rds`, `wind_speed.rds`, and the final `analytic_dataset.rds`—are written to disk alongside the audit artefacts (`dropped_hexes.csv` and the row/hex ledger log), so that any individual step can be re-run or independently checked without rebuilding the whole chain.

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